

Growth, Not Decay, for $E_{1,\beta}(i\lambda)$ When $0 < \beta < 1$

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Status. Candidate counterexample, likely valid, for human review.

1 Source question

Ruzhansky and Torebek study

$$I_{\alpha,\beta}(\lambda) = \int_a^b E_{\alpha,\beta}(i\lambda\phi(x))\psi(x) dx, \quad E_{\alpha,\beta}(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\alpha n + \beta)}.$$

Their theorem for $\alpha = 1$ and $\beta > 1$ assumes a real bounded phase, an L^1 amplitude, and $\inf |\phi| > 0$, and obtains decay of order $\lambda^{1-\beta}$. Open Question 1 asks [1, Conclusion, PDF p. 28]:

Are van der Corput-type estimates valid for $I_{\alpha,\beta}(\lambda)$ with $\alpha = 1$ and $\beta < 1$?

CONCLUSION AND SOME OPEN QUESTIONS

The main object of the paper was to estimate integral operator defined by

$$I_{\alpha,\beta}(\lambda) = \int_a^b E_{\alpha,\beta}(i\lambda\phi(x))\psi(x)dx,$$

with $0 < \alpha \leq 1$, $\beta > 0$.

For $I_{\alpha,\beta}(\lambda)$, the following results have been obtained:

- for different cases of parameters α and β , some analogues of the van der Corput first lemma are obtained;
- for different cases of parameters α and β , some analogues of the van der Corput second lemma are obtained;
- for $0 < \alpha \leq 1/2$ and $\beta \geq \alpha$ bilateral optimal bounds estimates are obtained;
- some applications of the generalised van der Corput lemmas are presented.

In conclusion, we present some open questions related to generalisations of the van der Corput lemmas:

- (1) are van der Corput-type estimates valid for $I_{\alpha,\beta}(\lambda)$ with $\alpha = 1$ and $\beta < 1$?

We interpret the broad question in the generality of the immediately preceding $\alpha = 1$ theorem: arbitrary bounded measurable phases separated from zero and L^1 amplitudes. In that class, the answer is negative for decay, and the sharp replacement scale is growth of order $\lambda^{1-\beta}$.

2 Claimed contribution

Throughout, complex powers use the principal branch.

Theorem 1 (Sharp nonvanishing-phase scale). *Fix $0 < \beta < 1$ and $0 < m \leq M < \infty$. Suppose $\phi : [a, b] \rightarrow \mathbb{R}$ is measurable,*

$$m \leq |\phi(x)| \leq M \quad a.e., \quad \psi \in L^1[a, b].$$

Then, as $\lambda \rightarrow +\infty$,

$$I_{1,\beta}(\lambda) = \lambda^{1-\beta} \int_a^b e^{i\lambda\phi(x)} (i\phi(x))^{1-\beta} \psi(x) dx + O_{\beta,m}(\lambda^{-1} \|\psi\|_1). \quad (1)$$

In particular,

$$|I_{1,\beta}(\lambda)| \leq C_{\beta,m,M} (1 + \lambda)^{1-\beta} \|\psi\|_1. \quad (2)$$

This scale is sharp: for $[a, b] = [0, 1]$ and $\phi \equiv 1$, the operator norm of $\psi \mapsto I_{1,\beta}(\lambda)$ on $L^1[0, 1]$ satisfies

$$\|I_{1,\beta}(\lambda)\|_{(L^1)^*} = |E_{1,\beta}(i\lambda)| = (1 + o(1)) \lambda^{1-\beta}. \quad (3)$$

Consequently, no uniform estimate with a decaying right-hand side is valid in this class.

3 Proof intuition

For $\beta > 1$, the source paper obtains decay because the oscillatory exponential hidden inside $E_{1,\beta}$ is multiplied by $\lambda^{1-\beta}$. Crossing below $\beta = 1$ reverses that prefactor: it grows. An exact beta-integral expresses $E_{1,\beta}$ through an incomplete gamma function. The standard incomplete-gamma expansion then shows that the growing exponential term has modulus exactly $\lambda^{1-\beta}$, while all remaining terms are only $O(\lambda^{-1})$. Taking the allowed constant phase $\phi = 1$ removes any integration cancellation, so the growth is actually attained. Formula (1) records what remains for nonconstant phases: a classical oscillatory integral multiplied by the growing $\lambda^{1-\beta}$ prefactor.

4 Proof

Lemma 1 (Imaginary-axis asymptotic). *For fixed $0 < \beta < 1$,*

$$E_{1,\beta}(it) = (it)^{1-\beta} e^{it} + \frac{1-\beta}{\Gamma(\beta)it} + O_\beta(t^{-2}) \quad (t \rightarrow +\infty). \quad (4)$$

The analogous estimate holds uniformly on the two rays $z = its$ with $m \leq |s| \leq M$.

Proof. Reindexing the defining series gives the exact recurrence

$$E_{1,\beta}(z) = \frac{1}{\Gamma(\beta)} + zE_{1,\beta+1}(z). \quad (5)$$

For $\beta > 0$, termwise integration of the exponential series gives

$$\begin{aligned} E_{1,\beta+1}(z) &= \frac{1}{\Gamma(\beta)} \int_0^1 e^{zs} (1-s)^{\beta-1} ds \\ &= z^{-\beta} e^z \frac{\gamma(\beta, z)}{\Gamma(\beta)}, \end{aligned} \quad (6)$$

where the last substitution follows the straight ray from 0 to z and uses a consistent branch. Since $\gamma(\beta, z) = \Gamma(\beta) - \Gamma(\beta, z)$, equations (5)–(6) yield

$$E_{1,\beta}(z) = \frac{1}{\Gamma(\beta)} + z^{1-\beta} e^z - \frac{z^{1-\beta} e^z \Gamma(\beta, z)}{\Gamma(\beta)}. \quad (7)$$

The standard fixed-parameter expansion for the upper incomplete gamma function is

$$\Gamma(\beta, z) = z^{\beta-1} e^{-z} \left(1 + \frac{\beta-1}{z} + O_\beta(z^{-2}) \right), \quad (8)$$

in every closed subsector of $|\arg z| < 3\pi/2$ [3, Sec. 8.11(i), Eqs. 8.11.2–8.11.3]. This includes both imaginary rays and is uniform when $z = its$ with $m \leq |s| \leq M$. Substituting (8) into (7) cancels the two constant terms and gives

$$E_{1,\beta}(z) = z^{1-\beta} e^z + \frac{1-\beta}{\Gamma(\beta)z} + O_\beta(z^{-2}),$$

which proves the lemma. □

Apply Lemma 1 pointwise with $z = i\lambda\phi(x)$. Uniformity and $|\phi| \geq m$ give

$$E_{1,\beta}(i\lambda\phi(x)) = \lambda^{1-\beta} (i\phi(x))^{1-\beta} e^{i\lambda\phi(x)} + O_{\beta,m}(\lambda^{-1}).$$

Multiplication by ψ and integration proves (1). Taking absolute values and using $|(i\phi)^{1-\beta}| = |\phi|^{1-\beta} \leq M^{1-\beta}$ proves (2) for large λ ; bounded λ are absorbed into the constant by continuity of the entire Mittag-Leffler function.

Finally, let $\phi \equiv 1$. Then

$$I_{1,\beta}(\lambda) = E_{1,\beta}(i\lambda) \int_0^1 \psi(x) dx.$$

The norm of the integration functional on $L^1[0, 1]$ is one, hence the operator norm is $|E_{1,\beta}(i\lambda)|$. By Lemma 1,

$$\left| E_{1,\beta}(i\lambda) - (i\lambda)^{1-\beta} e^{i\lambda} \right| = O(\lambda^{-1}).$$

The leading term has modulus $\lambda^{1-\beta}$, so division by that quantity proves (3). In particular the norm tends to infinity, which rules out every uniform decay estimate and completes the proof of Theorem 1.

Corollary 1 (Explicit counterexample). *For every $0 < \beta < 1$, the data*

$$[a, b] = [0, 1], \quad \phi(x) = 1, \quad \psi(x) = 1$$

satisfy the hypotheses of the nonvanishing-phase class, but

$$|I_{1,\beta}(\lambda)| \sim \lambda^{1-\beta} \longrightarrow \infty.$$

Thus the decay conclusion suggested by the source question is false.

5 Verification and limitations

The analytic audit is recorded in `VERIFIER_REPORT.md`. The script `code/verify_asymptotic.py` evaluates the exact lower-incomplete-gamma representation at 70-digit precision at the nine points

$$\beta \in \{0.2, 0.5, 0.8\}, \quad t \in \{30, 100, 300\}.$$

It checks both

$$\frac{|E_{1,\beta}(it)|}{t^{1-\beta}} \approx 1 \quad \text{and} \quad t \left| E_{1,\beta}(it) - (it)^{1-\beta} e^{it} \right| \approx \frac{1-\beta}{\Gamma(\beta)}.$$

These computations are sanity checks and are not used in the proof.

The theorem settles the no-decay question in the broad measurable, nonvanishing-phase class of the source’s adjacent $\alpha = 1$ result. It does not assert that all derivative-restricted estimates fail. Indeed, formula (1) shows that under smooth phase hypotheses the problem becomes a classical oscillatory integral estimate for the modified amplitude $(i\phi)^{1-\beta}\psi$, multiplied by $\lambda^{1-\beta}$. The main interpretive review point is whether this is the intended scope of the source’s unqualified phrase “van der Corput-type estimates.”

6 Novelty check

The bounded search on 21 July 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; the exact arXiv id and title; the verbatim open question; searches combining $E_{1,\beta}$, $\beta < 1$, Mittag-Leffler, and van der Corput; and the authors’ names with possible later parts. We checked the full local sources of Part I and Part II. Part II [2] treats $\beta > 1$ and the borderline $\beta = 1$, but not $\beta < 1$. No explicit later answer was found. This bounded search supports, but cannot certify, novelty.

Human review recommendation. Send to a special-functions or oscillatory-integrals reviewer. The key checks are the branch convention in Lemma 1 and the scope interpretation just stated.

References

- [1] M. Ruzhansky and B. T. Torebek, *Van der Corput lemmas for Mittag-Leffler functions. I*, arXiv:2002.07492 (2020; revised 2024), <https://arxiv.org/abs/2002.07492>.

- [2] M. Ruzhansky and B. T. Torebek, *Van der Corput lemmas for Mittag-Leffler functions. II. α -directions*, arXiv:2005.04546 (2020), <https://arxiv.org/abs/2005.04546>.
- [3] NIST Digital Library of Mathematical Functions, *Incomplete Gamma Functions: Asymptotic Approximations and Expansions*, Sec. 8.11(i), Eqs. 8.11.2–8.11.3, <https://dlmf.nist.gov/8.11>.